



Introduction

Electrical & Electronic Circuits and Elements

Embedded Systems & Embedded Programming

Fundamentals of Electricity

❖ **Electricity** is briefly defined as the flow of **electric charges**

❖ **Electric charge (Q)**

- Is a fundamental property of matters, like mass
- Causes matters to experience a force when they are placed in an electromagnetic field
- The smallest amount of charge that exists is the charge carried by an electron ($1.602 \times 10^{-19} \text{ C}$)
- Is measured in coulombs (C)
 - A coulomb is defined as the **quantity of charge** that flows through the cross section of an electrical conductor carrying one **ampere** for one **second**. $1 \text{ C} = 6.241 \times 10^{18}$ charges
- Exists in **positive** (+) or **negative** (-) forms

Fundamentals of Electricity (Charge)

❖ Charges are carried by electrons and protons

➤ $q_e = -1.602 \times 10^{-19} \text{ C}$ and $q_p = +1.602 \times 10^{-19} \text{ C}$

❖ Electrostatic force (**Coulomb's law**)

➤ Is the force that operates between charges

➤ Opposites attract and likes repel

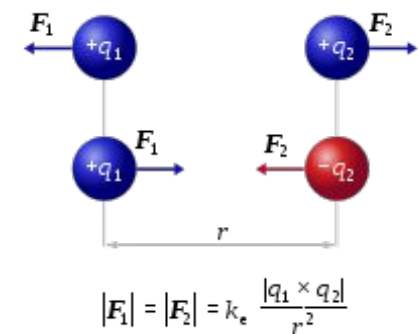
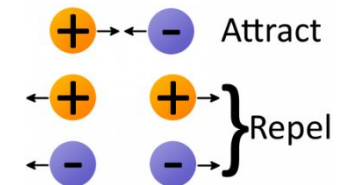
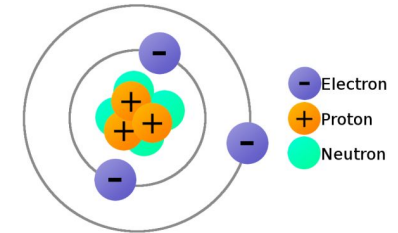
■ e.g. electrons push away other electrons and are attracted by protons

➤ The amount of the force depends on how far they are from each other

➤ The closer two charges get, the greater the force becomes

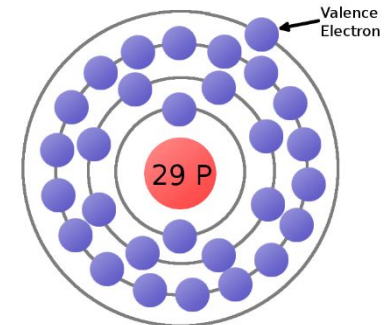
➤ Is one of those forces that hold atoms together

➤ But it can also make electrons (and charges) flow!!

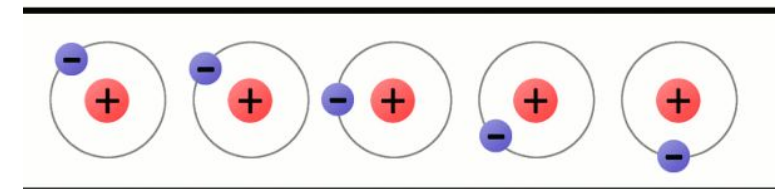


Fundamentals of Electricity (Charge Flow)

- ❖ The outermost electrons of an atom are called the **valence** electrons
- ❖ **Valence** electrons require the least amount of force to be freed from an atom
- ❖ By applying enough electrostatic force to a valence electron we can free it
 - Pushing it by another negative charge
 - Attracting it by a positive charge
- ❖ A free electron in the space between atoms
 - Eventually finds a new atom to latch on to
 - And its negative charge pushes another valence electron from the atom
 - This chain effect creates the flow of electrons (charges)
 - And the flow of charges is **CURRENT**



The Bohr model of an atom of copper



Fundamentals of Electricity (Conductivity)

- ❖ Some types of atoms are better than others at releasing their electrons
- ❖ Conductivity measures how tightly bound an electron is to an atom
- ❖ Conductors
 - Elements with high conductivity, which have very mobile electrons
 - Metals like copper, silver, and gold etc.
- ❖ Insulators
 - Elements with low conductivity that prevent the flow of electrons
 - Popular insulators include glass, rubber, plastic, and air
- ❖ Semiconductors
 - Have a conductivity between conductors and insulators
 - Let electrons flow under certain conditions



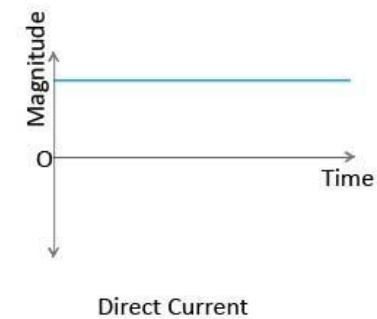
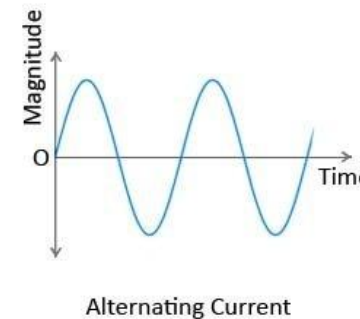
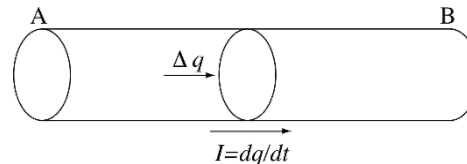
[What is Electric Charge and How Electricity Works](#)

Fundamentals of Electricity (Current)

- ❖ In order to flow charges, current, a circuit is required
- ❖ A circuit is a closed, never-ending loop of conductive material
- ❖ Current is defined as the time rate of flow of charges through a predetermined area of a conductor (Amount of charge per second)
- ❖ Current unit is **ampere** or **C/sec**
- ❖ Can be measured by Multimeters (series connected)
- ❖ It can be in AC (Alternating Current) or DC (Direct Current) forms
 - DC current has a constant value
 - AC current has a value that changes sinusoidally

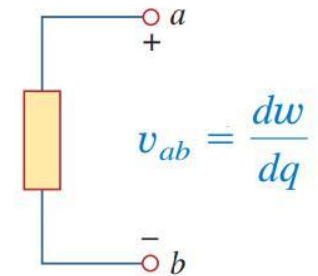


[Basic Electricity - What is an amp?](#)



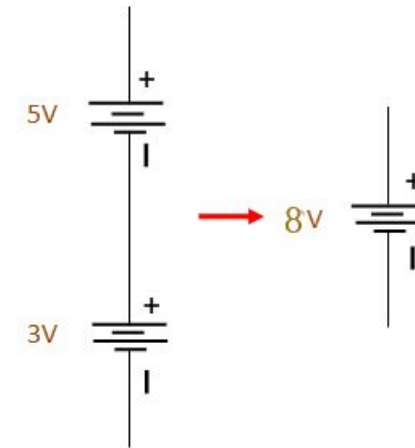
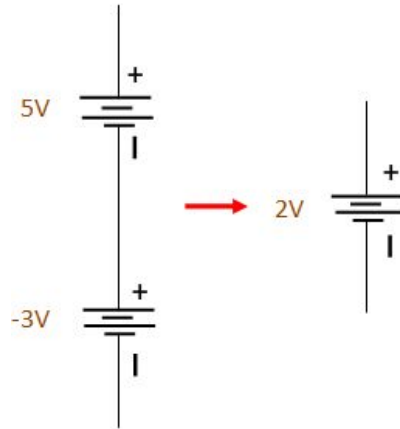
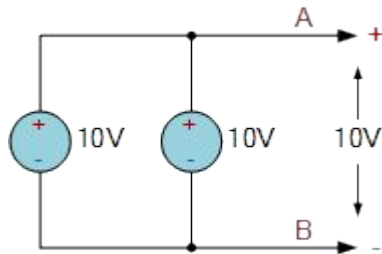
Fundamentals of Electricity (Voltage)

- ❖ Moving charge in a direction between two points in a circuit, takes some energy
- ❖ **Voltage** (potential difference) is the energy required to move a unit of charge through an element
 - Measured in **volts (V)** or **joule (J) / coulomb (C)**
- ❖ A **voltage source** raises the electrical potential of the charged particles
 - So charges can move in the circuit
- ❖ A constant voltage is called a **DC voltage** and is represented by **V**
- ❖ A sinusoidally time-varying voltage is called an **AC voltage** and is represented by **v**
- ❖ Voltage is always relative
 - Is measured between two points
 - A point in the circuit should be used as **reference voltage** (like ground; **GND**)



Fundamentals of Electricity (Voltage)

- ❖ It can be measured by multimeters and oscilloscopes (parallel connected)
- ❖ Series connection of voltage sources
- ❖ Parallel connection of voltage sources
- ❖ Polarities matter



[Basic Electricity - What is voltage?](#)

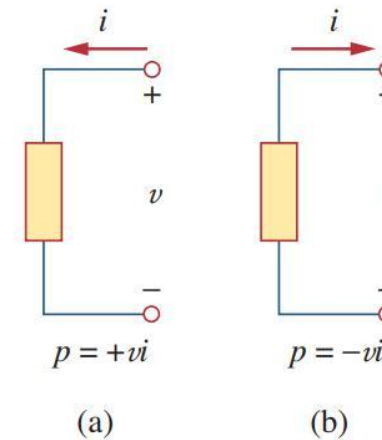
Fundamentals of Electricity (Power)

- ❖ Power is the time rate of generating or consuming energy
- ❖ Measured in **watts (W)** or **joules (J) / seconds (s)**

$$p = \frac{dw}{dt}$$

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

$$p = vi$$



Reference polarities for power using the passive sign convention: (a) absorbing power, (b) supplying power.



[Basic Electricity - Power and watts](#)

Electrical and Electronic Circuits/ Devices (Differences)

Electrical Circuits & Devices	Electronic Circuits & Devices
Convert the electrical current into another form of energy like heat, light, etc.	Control the electrical current to perform a task
Use copper and aluminum wires for the flow of current	Use semiconductor materials like silicone
Mainly use alternating current (AC)	Use direct current (DC)
Work on high voltages	Work on low voltages
High power consumption	Low power consumption
High conductivity	Low conductivity
Do not process information	Process and store information

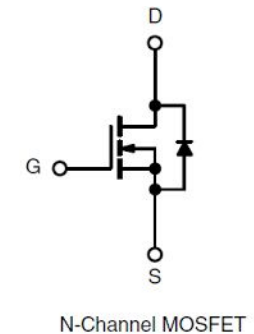
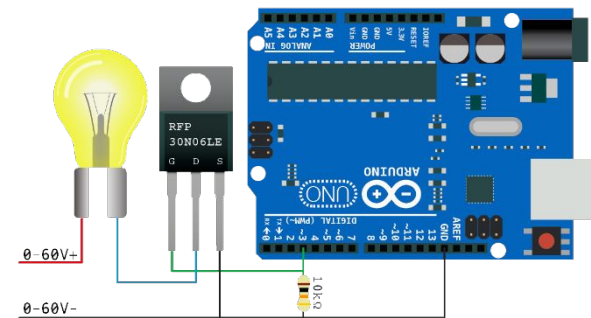
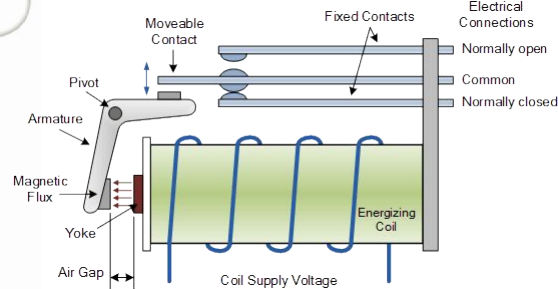
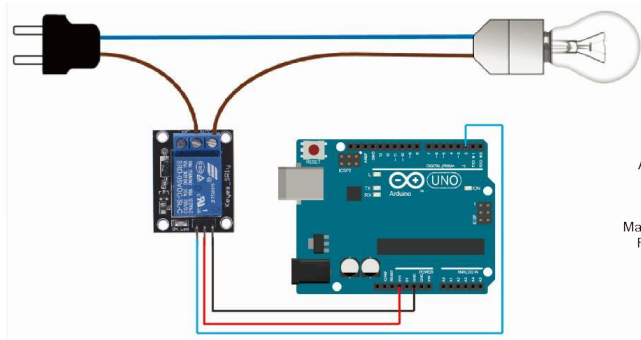
Electrical and Electronic Circuits/ Devices (Differences)

Electrical Circuits & Devices	Electronic Circuits & Devices
Operate on analog signals	Operate on digital signals
Deal with kA, kV and MW	Deal with mA, mV and mW
Usually heavy and larger in size	Very small and may contain thousands of tiny components
Use passive elements ¹ like resistors, capacitors and inductors	Use active elements ² like transistors, battery, diodes and amplifiers
Usually operate in low frequencies	Usually operate in high frequencies
More dangerous if short circuit occurs	Less dangerous

1. Elements incapable of controlling current by means of another electrical signal
2. Elements with ability to electrically control current or supplying energy to the circuit

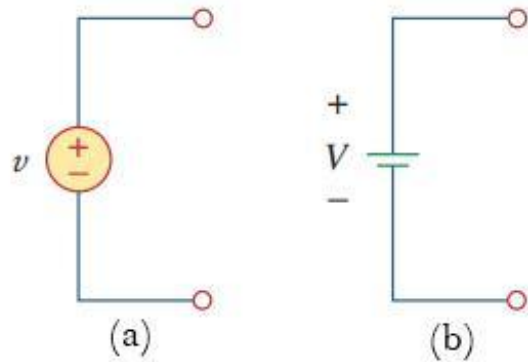
Electrical and Electronic Circuits/ Devices (Similarities)

- ❖ Both use the electrical energy for doing work
- ❖ Both are interrelated with each other
- ❖ The link between them is typically provided by relays or transistors
- ❖ Briefly, all electronic devices are also electrical devices
- ❖ Both are manufactured in commercial, industrial and military categories

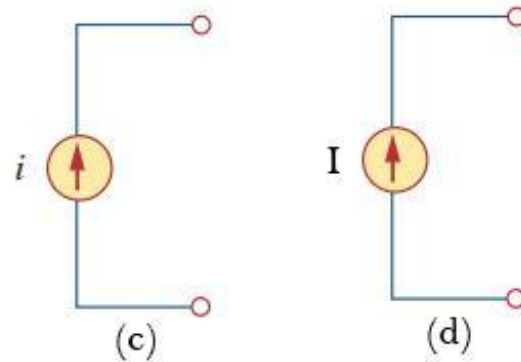


Fundamentals of Electric Circuits (Definitions)

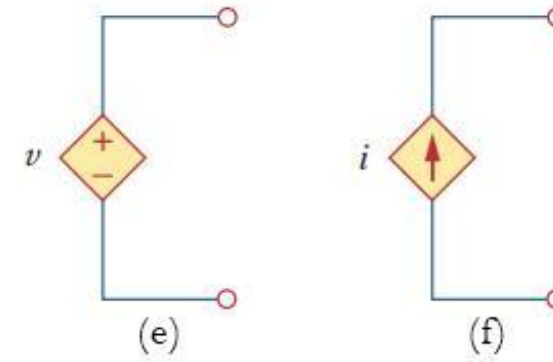
- ❖ **Ideal independent source:** An active element that provides a specified voltage or current that is completely independent of other circuit elements
- ❖ **Ideal dependent source:** An active element that is controlled by another source



General symbol for ideal independent voltage source.
(a) time-varying voltage
(b) constant voltage (DC)



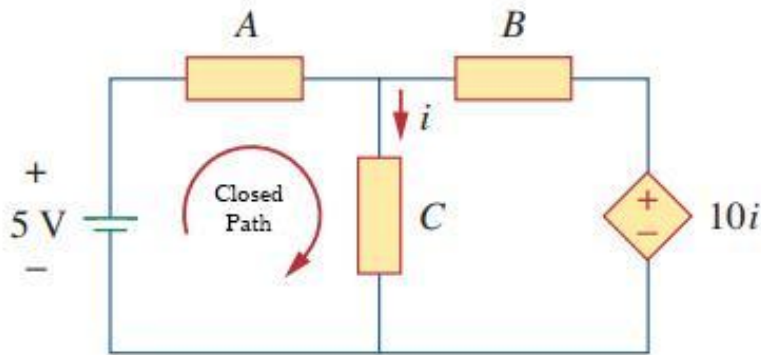
General symbol for ideal independent current source.
(c) time-varying current
(d) constant current (DC)



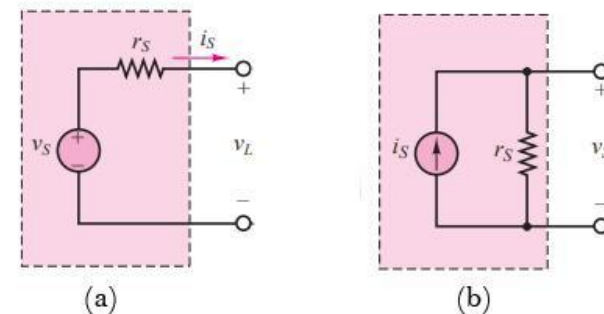
General symbol for ideal dependent source.
(e) dependent voltage
(f) dependent current

Fundamentals of Electric Circuits (Definitions)

- ❖ **Practical Sources:** Because of the internal resistor of a voltage/current source the delivered power is not infinite and depends on the load.
- ❖ **Network:** Topology of interconnection of electrical elements or devices
 - Study the properties relating to the placement of elements in the network
- ❖ **Circuit:** A network providing one or more closed paths



Network and circuit example
Dependent & Independent Sources



Practical Source Models
(a) Voltage Source
(b) Current Source

Fundamentals of Electric Circuits (Definitions)

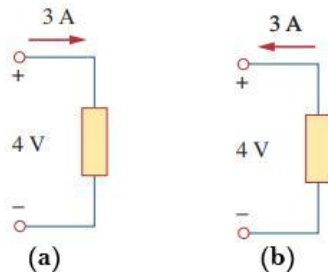
❖ The law of conservation of energy

- In a circuit, at any instant of time, **consumed power + supplied power = 0**, ($\sum p = 0$)

❖ Current direction and voltage polarity play a major role in circuit analysis

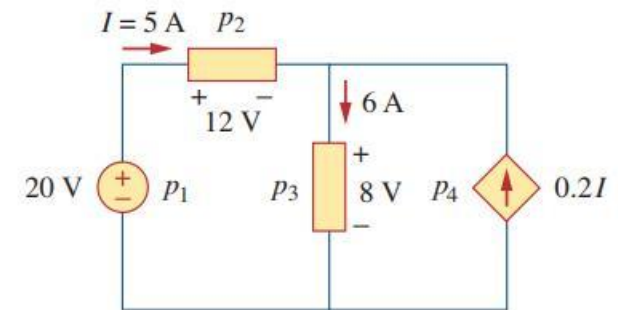
❖ Passive sign convention

- Positive current enters to the positive terminal
- If power ($v \cdot i$) is greater than 0 then the element is consuming power
- If power ($v \cdot i$) is lesser than 0 then the element is supplying power
- For passive elements, current always flows into the positive terminal



Reference polarities for power using the passive sign convention
(a) Absorbing power, 12 watt
(b) Supplying power, -12 watt

$P_1 = -100 \text{ W}$ (Supplied power)
 $P_2 = 60 \text{ W}$ (Absorbed power)
 $P_3 = 48 \text{ W}$ (Absorbed power)
 $P_4 = -8 \text{ W}$ (Supplied power)



Fundamentals of Electric Circuits (Definitions)

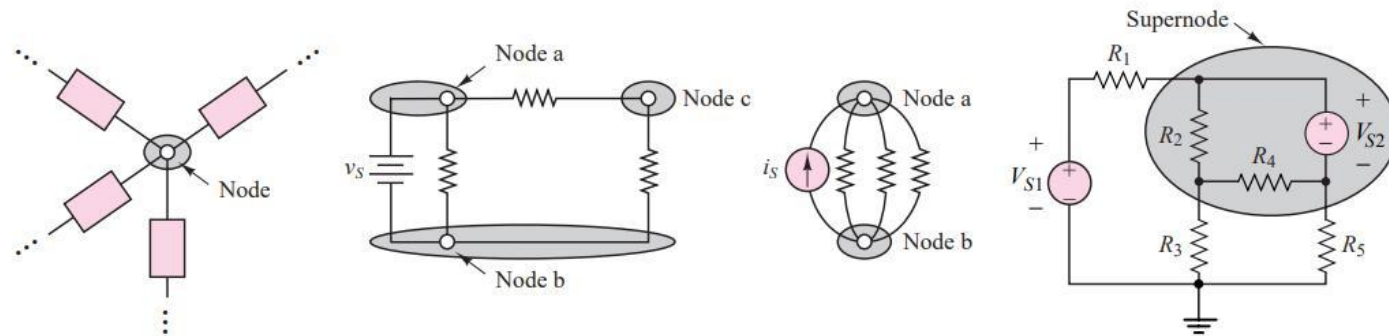
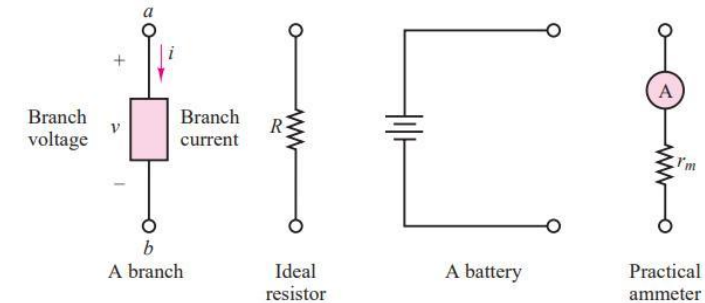
❖ Branch

- Any portion of a circuit with two terminals
- May consist of one or more circuit elements

❖ Node: The junction of two or more branches

❖ Super node

- A region that encloses more than one node
- Can be treated in the same way as nodes

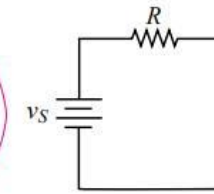
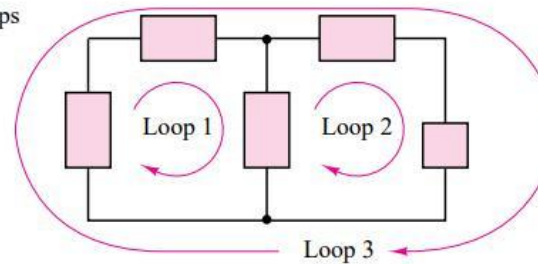


Fundamentals of Electric Circuits (Definitions)

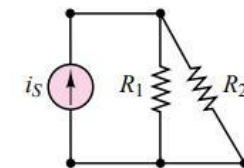
❖ Loop

- Any closed connection of branches

Note how two different loops in the same circuit may include some of the same elements or branches.



1-loop circuit

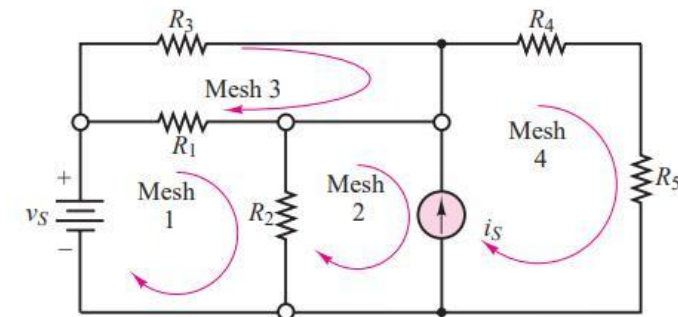


3-loop circuit
(How many nodes in this circuit?)

❖ Mesh

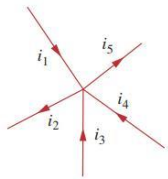
- A loop that does not contain other loops

How many loops can you identify in this four-mesh circuit? (Answer: 15)

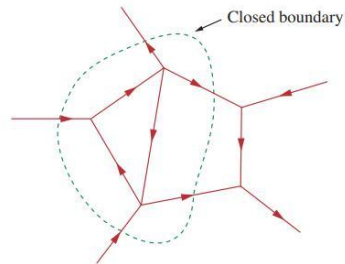


Fundamentals of Electric Circuits (KCL)

- ❖ In order to flow current, there must exist a closed circuit
- ❖ The law of conservation of charge
 - Charge cannot be created or destroyed
- ❖ Kirchhoff's current law (KCL)
 - The algebraic sum of currents at a node (or a closed boundary) is zero
 - May be interpreted as **total entering currents = total leaving currents**



Currents at a node illustrating KCL.



Applying KCL to a closed boundary.

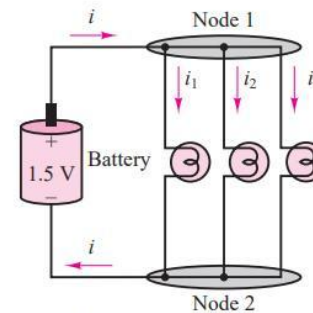


Illustration of KCL at node 1: $-i + i_1 + i_2 + i_3 = 0$

$$\sum_{n=1}^N i_n = 0 \quad \text{Kirchhoff's current law}$$

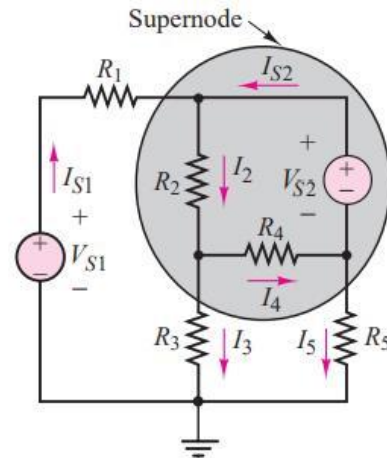
Fundamentals of Electric Circuits (KCL)

❖ Application of KCL (example)

If $I_2 = 3\text{ A}$, $I_3 = 2\text{ A}$, $I_4 = 1\text{ A}$,
 $I_5 = 0\text{ A}$, find I_{S1} and I_{S2}

Answer:

$I_{S1} = 2\text{ A}$, $I_{S2} = 1\text{ A}$

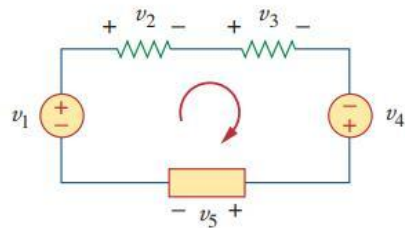


[Kirchhoff's current law](#)

Fundamentals of Electric Circuits (KVL)

- ❖ The law of conservation of energy
 - Energy is not lost or created in an electric circuit
- ❖ Kirchhoff's voltage law (KVL)
 - The algebraic sum of all voltages around a closed path (or loop) is zero
 - May be interpreted as **Sum of voltage drops = Sum of voltage rises**

KVL can be applied in two ways: by taking either a clockwise or a counterclockwise trip around the loop. Either way, the algebraic sum of voltages around the loop is zero.



A single-loop circuit illustrating KVL.

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

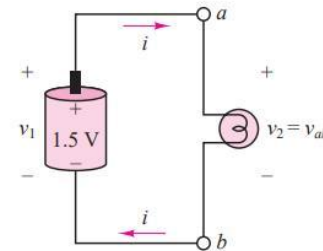


Illustration of Kirchhoff's voltage law: $v_1 = v_2$

$$\sum_{n=1}^N v_n = 0 \quad \text{Kirchhoff's voltage law}$$

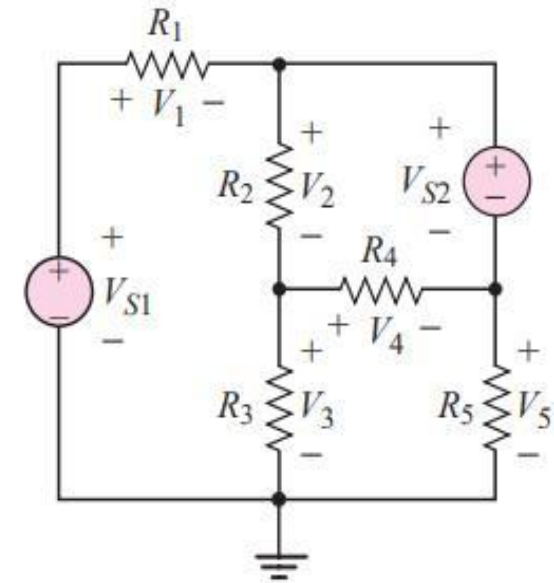
Fundamentals of Electric Circuits (KVL)

❖ Application of KVL (example)

If $V_{S1} = 12\text{ V}$, $V_{S2} = -4\text{ V}$, $V_2 = 2\text{ V}$,
 $V_3 = 6\text{ V}$, $V_5 = 12\text{ V}$, find V_1 , V_4 .

Answer: By applying KVL clockwise
around each of the three meshes:

$V_1 = 4\text{ V}$ and $V_4 = -6\text{ V}$



Fundamentals of Electricity

❖ References

- Fundamentals of Electric Circuits, 5th Edition, Charles K. Alexander and Matthew N.O. Sadiku, ISBN 978-0-07-338057-5
- Fundamentals of Electrical Engineering, First Edition, Giorgio Rizzoni, ISBN 978-0-07-338037-7
- Most of the images have been copied from these books

❖ Some useful links

- [What is Electricity?](#)
- [Coulomb's Law | Electronics Basics](#)
- [Electric Field | Electronics Basics](#)
- [Electric Potential and Electric Potential Difference \(Voltage\)](#)
- [Electric Circuits: Basics of the voltage and current laws](#)
- [Battery Energy and Power](#)
- [Sign convention for passive components and sources](#)
- [Differences Between Active and Passive Elements](#)

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a